



University of Colombo, Sri Lanka
University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination - Semester I - UCSC AY21 [held in Sept / Oct 2024]

SCS1304 — Problem Solving Strategies and Computational Approaches

(Two (2) Hours)

Answer ALL questions

Number of Pages = 22

Number of Questions = 3

To be completed by the candidate

Index Number:

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Important Instructions to Candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number CLEARLY on each and every page of this Question paper.
- IV. This paper consists of three (03) questions in 22 pages (including the Cover Page).
- V. Answer ALL questions.
- VI. Programmable Calculators and any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, are **not allowed**.
- VII. Non-Programmable calculators are also **not allowed**
- VIII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate.
- IX. The first question carries 40 marks, and the second and third questions carry 30 marks each.

To be completed by the examiners

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2	
3	
Total	

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Q1.	i. Question 1 has 20 MCQ-based questions. ii. Each question carries 1 mark. iii. Each MCQ question will have 5 (five) choices with one correct answer. iv. Select the most correct answer. v. Answers should be marked on given MCQ Sheet (provided separately) (40 marks)
1)	Which of the following shows the correct order of time complexity?
	(a) $O(n \log n) < O(n) < O(\log n) < O(n^2) < O(n^3) < O(n!) < O(2^n)$
	(b) $O(n \log n) < O(n) < O(\log n) < O(n^2) < O(n^3) < O(2^n) < O(n!)$
	(c) $O(\log n) < O(n) < O(n \log n) < O(n^2) < O(n^3) < O(2^n) < O(n!)$
	(d) $O(n) < O(\log n) < O(n \log n) < O(n^2) < O(n^3) < O(2^n) < O(n!)$
	(e) $O(n) < O(\log n) < O(n \log n) < O(n^2) < O(n^3) < O(n!) < O(2^n)$
2)	The effect of doubling the input size on an algorithm with time complexity $O(n^3)$ is
	(a) Two times
	(b) Three times
	(c) Four times
	(d) Eight times
	(e) Negligible
3)	<i>Greedy Choice Property</i>
	(a) reaches the solution (the global optimum) by making locally optimal choices.
	(b) involves finding a solution incrementally by trying different options and undoing.
	(c) reconsiders previous choices to improve the solution.
	(d) makes the best global choice at each stage.
	(e) always divide the problem into equal parts.
4)	Which of the following algorithm is most suitable to solve by using <i>Dynamic Programming</i> ?
	(a) Selection Sort
	(b) Binary Search
	(c) Merge Sort
	(d) Quick Sort
	(e) Longest Common Sub-sequence

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5)	What is the time complexity of calculating the <i>Fibonacci Sequence</i> using <i>Dynamic Programming</i> ?	
	(a)	$O(n)$
	(b)	$O(2^n)$
	(c)	$O(\log n)$
	(d)	$O(n \log n)$
	(e)	$O(n^2)$
6)	Which of the following is/are the correct difference(s) between the bottom-up approach (<i>Tabulation</i>) and the top-down approach (<i>Memorization</i>) in <i>Dynamic Programming</i> ?	
	A. In general, <i>Tabulation</i> builds from small to large subproblems, while <i>Memorization</i> works from large to small.	
	B. <i>Tabulation</i> can be slow due to recursive calls and return statements.	
	C. Code can be less complicated in <i>Memorization</i> compared to <i>Tabulation</i> .	
	(a)	A only
	(b)	A and B only
	(c)	B and C only
	(d)	A and C only
	(e)	All A,B and C
7)	Consider the following two sequences; S1 = {B, C, D, A, A, C, D} S2 = {A, C, D, B, A, C} What is the <i>Longest Common Subsequence (LCS)</i> of S1 and S2?	
	(a)	{A, A, C}
	(b)	{D, A, C}
	(c)	{B, A, C}
	(d)	{C, D, A, B}
	(e)	{C, D, A, C}
8)	How many guesses would you need to determine the number out of 1 through 512 in the worst case using a <i>Binary Search</i> algorithm?	
	(a)	6
	(b)	7
	(c)	8
	(d)	9
	(e)	10

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9)	Which of the following statement(s) is/are correct regarding <i>Strassen's Algorithm</i> for multiplying matrices using <i>Divide and Conquer</i> approach? A. The algorithm for matrices of dimension n has a total number of scalar arithmetic and subtraction complexity of $6n^{2.81} - 6n^2$ B. <i>Strassen's algorithm</i> for multiplying matrices requires seven multiplications and 18 additions C. <i>Strassen's algorithm</i> uses a divide and conquer approach	
	(a)	A only
	(b)	B only
	(c)	C only
	(d)	A and C only
	(e)	All A,B and C
10)	Consider the following pseudocode that recursively calculates the <i>factorial</i> of a number n with missing terms P and Q; <pre> Integer factorial(n){ if n == 0: return 1 returnP... * ...Q.... } </pre> Which of the following correctly replace the labels P and Q above?	
	(a)	P -> $n-1$, Q -> factorial(n)
	(b)	P -> n , Q -> factorial(n)
	(c)	P -> n , Q -> factorial($n - 1$)
	(d)	P -> n , Q -> factorial($n - 2$)
	(e)	P -> 1 , Q -> factorial($n - 1$)
11)	Assume that you have an infinite number of coins of denominations (Values) {1, 2, 5}. How many coins will be used to make 28 using a <i>Greedy Algorithm</i>? Assume that the <i>Greedy algorithm</i> works optimally.	
	(a)	5
	(b)	6
	(c)	7
	(d)	8
	(e)	9

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12)	Consider the following statements regarding <i>Quick Sort</i>; A. Best Case Ω of Quick Sort for input size n is $\log n$. B. The best-case scenario for quicksort occurs when the pivot chosen at each step divides the array into roughly equal halves. C. Worst Case O of Quick Sort is when the array is already sorted, and the pivot is always chosen as the smallest or largest element. Which of the above statement(s) is/are correct?	
	(a)	A only
	(b)	B only
	(c)	A and C only
	(d)	B and C only
	(e)	All A,B and C
13)	What is the time complexity of the following? <pre> total = 0; for (int i = n/2 ; i < n ; i++) { for (int j = n/2 ; j < n ; j++) { for (int k = 0 ; k < n ; k++) { total = total + j } } } </pre>	
	(a)	$O(n^2)$
	(b)	$O(n^3)$
	(c)	$O(\log n)$
	(d)	$O(n \log n)$
	(e)	$O(n^2 \log n)$
14)	Which of the following statements is incorrect regarding recursion?	
	(a)	A recursive function calling another instance of the same function for one time is known as <i>Linear Recursion</i>
	(b)	A recursive function calling another instance of the same function for more than one time is known as <i>Tree Recursion</i>
	(c)	A recursive function which passes the parameter as a recursive call is called <i>Nested Recursion</i>
	(d)	A recursive function calling another instance of the same function and that recursive call is the first statement in the function is known as <i>Head Recursion</i>
	(e)	A function is called within another instance of the same function directly is called <i>Indirect Recursion</i>

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15)	The following four steps, P,Q,R, and S, outline the <i>Four Queens Problem Algorithm</i> using Backtracking. Note that they are not in the correct order. P - While placing a queen in a row, check for clashes with already placed queens. Q - In case, if no safe cell found due to clashes, then backtrack (i.e, undo the placement of recent queen) and return false. R - For any column, if there is NO clash then mark this row and column as part of the solution by placing the queen. S - Place each queen one by one in different rows, starting from the top most row. Which of the following is the correct order of sequence of steps?	
	(a)	P,Q,R,S
	(b)	S,Q,R,P
	(c)	Q,P,S,R
	(d)	P,R,Q,S
	(e)	S,R,Q,P
16)	The Worst case occur in linear search algorithm when _____	
	(a)	Item is somewhere in the middle of the array
	(b)	Item is not in the array at all
	(c)	Item is the last element in the array
	(d)	Item is the last element in the array or is not there in the array
	(e)	None of the above
17)	What is the time complexity of a function $3n^2+20n+5$?	
	(a)	$O(n)$
	(b)	$O(n \log n)$
	(c)	$O(\log n)$
	(d)	$O(n^2)$
	(e)	None of the above

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18)	What is the time complexity of following? <pre>total=0; for(int i = 1 ; i <= n ; i *= 2) total++;</pre>	
	(a)	$O(n)$
	(b)	$O(\log(n))$
	(c)	$O(n \log(n))$
	(d)	$O(n^2)$
	(e)	None of the above
19)	Which statement is true for large values of n ?	
	(a)	$(n^2 + n) / 2$ behaves like $n/2$
	(b)	$(n^2 + n) / 2$ behaves like n
	(c)	$(n^2 + n) / 2$ behaves like n^2
	(d)	$(n^2 + n) / 2$ behaves like $\log n$
	(e)	None of the above
20)	What does it mean when we say that algorithm P is <i>asymptotically</i> more efficient than Q ?	
	(a)	P will always be a better choice for small inputs
	(b)	P will always be a better choice for large inputs
	(c)	Q will always be a better choice for small inputs
	(d)	P will always be a better choice for all inputs
	(e)	Q will always be a better choice for large inputs

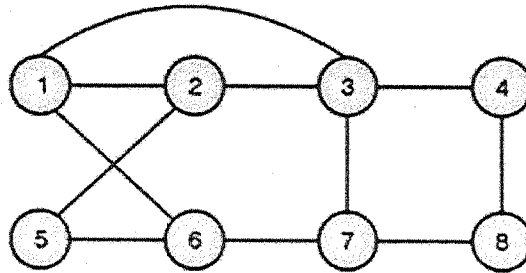
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Q2. A. What are the key pillars of *Computational Thinking*? Provide an example for each. (06 marks)

ANSWER BOX

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- Q2.** B. The **Hamiltonian Cycle** is the cycle in the graph that visits all the vertices exactly once and terminates at the starting node. The **backtracking** approach uses a **state-space tree** to check if there exists a **Hamiltonian cycle** in the graph. Draw a State Space Tree showing backtracking and find out **Hamiltonian Cycle(s)** for the following graph starting at **node 1**. State any assumptions made.



(07 marks)

ANSWER BOX

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- Q2.** C. The *divide-and-conquer* approach can be used in the *Quick Sort* algorithm. Draw and illustrate how a human would sort the following array using the *Quick Sort* algorithm. Assume that the **pivot** is chosen as the **last item**.

(7 marks)

19	7	15	12	16	18	4	11	13
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ANSWER BOX

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Q2.	D.	(i) Write down a recursive pseudocode to calculate <i>Fibonacci</i> terms. (ii) Draw the recursion tree to calculate <i>Fibonacci</i> term five (5). (10 marks) <u>ANSWER BOX</u>

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- Q3. A. (i) Study the following pseudo-code for *Quick Sort Algorithm* and write the relevant terms for labels P, Q, R, S and T in the given spaces;
- (ii) What are the *Worst*, *Average*, and *Best Case* time complexities (you are not required to derive the time complexity)? State any assumptions made.

```

void quicksort(int array[], int low, int high);
int partition(int array[], int low, int high);

int main() {
    int myArray[] = {64, 34, 25, 12, 22, 11, 90, 5};
    int n = sizeof(myArray) / sizeof(myArray[0]);

    quicksort(myArray, 0, n - 1);

    printf("Sorted array: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", myArray[i]);
    }
    return 0;
}

void quicksort(int array[], int low, int high) {
    if (low < high) {
        int pivotIndex = partition(array, low, high);
        quicksort(array, .....P....., .....Q..... );
        quicksort(array, .....R....., .....S..... );
    }
}

int partition(int array[], int low, int high) {
    int pivot = array[high];
    int i = low - 1;

    for (int j = low; j < high; j++) {
        if (array[j] <= ....T...) {
            i++;
            int temp = array[i];
            array[i] = array[j];
            array[j] = temp;
        }
    }

    int temp = array[i + 1];
    array[i + 1] = array[high];
    array[high] = temp;
    return i + 1;
}

```

(08 marks)

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ANSWER BOX

(i)

P:

Q:

R:

S:

T:

(ii)

Worst Case:

Best Case:

Average Case:

Index No.

- Q3. B. Consider the following adjacency matrix to be used in a *Branch and Bound Least Cost* approach in the *Travelling Salesman Problem (TSP)* algorithm. Note that N_0 to N_4 as nodes of the graph.

	N_0	N_1	N_2	N_3	N_4
N_0	INF	20	30	10	11
N_1	15	INF	16	4	2
N_2	3	5	INF	2	4
N_3	19	6	18	INF	3
N_4	16	4	7	16	INF

Starting from N_0 , calculate the relevant cost matrices and find out the cost of root nodes N_0 and N_1 in the *state space tree*. Fill in the spaces given as appropriate and state any assumptions made.

(7 marks)

ANSWER BOX

Calculate the cost of root node N_0 by filling the spaces given below.

Row Reduced Matrix

	N_0	N_1	N_2	N_3	N_4
N_0					
N_1					
N_2					
N_3					
N_4					

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Column Reduced Matrix

	N_0	N_1	N_2	N_3	N_4
N_0					
N_1					
N_2					
N_3					
N_4					

The Cost of node N_0 is =

.....

Calculate the cost of Node N_1 by filling the spaces given below.

Now add the edge from N_0 to N_1 in the *state space tree* and write down the resultant cost matrices below.

	N_0	N_1	N_2	N_3	N_4
N_0					
N_1					
N_2					
N_3					
N_4					

Index No.

Row Reduced Matrix

	N_0	N_1	N_2	N_3	N_4
N_0					
N_1					
N_2					
N_3					
N_4					

Column Reduced Matrix

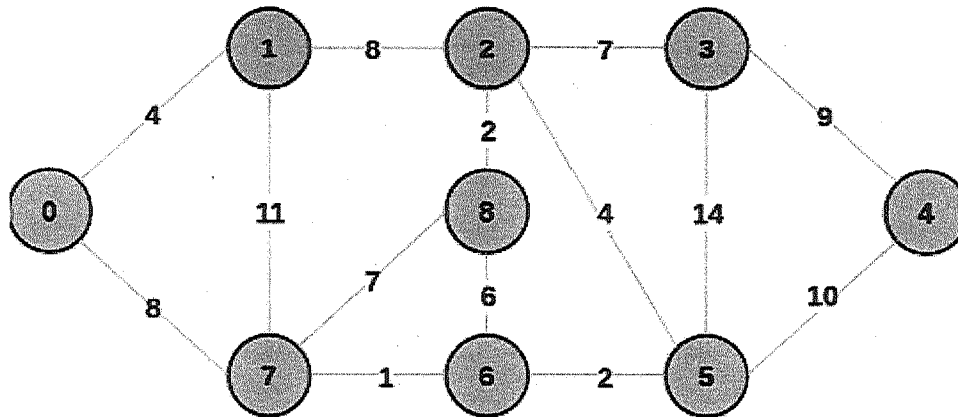
	N_0	N_1	N_2	N_3	N_4
N_0					
N_1					
N_2					
N_3					
N_4					

The Cost of node N_1 is =

.....

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Q3. C. *Prim's algorithm* is a *Greedy Algorithm* that finds a Minimum Spanning Tree (MST) for a weighted undirected graph. Show steps and draw the MST(s) and the total weight(s) of edges for the following graph using **Prim's algorithm**. State any assumptions made.



(08 marks)

ANSWER BOX

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Q3 D	<i>Kruskal's algorithm</i> is a greedy algorithm for finding the <i>Minimum Spanning Tree (MST)</i> of a graph. Show steps and draw the <u>MST</u> and the relevant <u>cost</u> of the following graph using <i>Kruskal's algorithm</i>. State any assumptions made. (07 Marks)
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ANSWER BOX

MST

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The Cost of MST:

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